

Equality Constrained Minimization

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1 Introduction

In this lecture, we begin the discussion of the methods for solving a convex optimization problem with equality constraints. We want to minimize $f(x)$,

$$\begin{aligned} \min f(x) \\ \text{s.t. } Ax = b, \end{aligned}$$

where $A \in \mathbb{R}^{p \times n}$.

To compare this problem with the general framework discussed in the previous lectures, we can rewrite the problem as follows:

$$\begin{aligned} \min f(x) \\ \text{s.t. } Ax - b = 0, \end{aligned}$$

where $A = \begin{bmatrix} a_1^T \\ a_2^T \\ \vdots \\ a_p^T \end{bmatrix}$ and $a_i, i \in \{1, 2, \dots, p\}$ are column vectors.

Then

$$Ax - b = \begin{bmatrix} a_1^T x - b_1 \\ a_2^T x - b_2 \\ \vdots \\ a_p^T x - b_p \end{bmatrix} = 0. \quad (1)$$

From Eqn. (1) we can determine the equality constraint h_j :

$$h_j(x) = a_j^T x - b_j,$$

for $j \in \{1, 2, \dots, p\}$

2 Convex Optimization

We are working with the convex optimization problem. To find a solution, make the following assumptions:

1. $f(x) : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex and twice continuously differentiable

2. $A \in \mathbb{R}^{p \times n}$ has full row rank, i.e. it has linearly independent rows and $\text{rank}(A) = p < n$
3. Optimal solution $x^* \in \text{dom } f$ exists and $p^* = f(x^*)$ will produce the smallest value of $f(x)$.

To solve this problem, try using the KKT conditions.

The Lagrangian without the inequality constraint λ is:

$$\begin{aligned}
 L(x, \nu) &= f(x) + \sum_{j=1}^p \nu_j h_j(x) \\
 &= f(x) + \sum_{j=1}^p \nu_j (a_j^T x - b_j) \\
 &= f(x) + \nu^T (Ax - b) \\
 &= f(x) + ((A^T \nu)^T x - \nu^T b),
 \end{aligned} \tag{2}$$

where $\nu \in \mathbb{R}^p$ is associated with the equality constraints.

The KKT conditions are as follows:

1. $\nabla_x L(x, \nu) = \nabla f(x) + A^T \nu = 0$, from (2)
2. NA since there are no inequality constraints,
3. $Ax = b$, from $h_j(x) = 0$, where $h_j = Ax - b$
4. NA since there are no inequality constraints,
5. NA since there are no inequality constraints.

Then (x^*, ν^*) satisfies

$$\begin{cases} \nabla f(x^*) + A^T \nu^* = 0 \\ Ax^* = b. \end{cases} \tag{3}$$

Since $x \in \mathbb{R}^n$ and $A \in \mathbb{R}^{p \times n}$,

$$\nabla f(x^*) + A^T \nu^* = 0$$

has n equations, and

$$Ax^* = b$$

has p equations.

Therefore, there are $n + p$ equations in the $n + p$ variables x^*, ν^* .

3 Example

Consider a simple example to build intuition.

10.1.1. Equality constrained convex quadratic minimization.

Let $f(x) = \frac{1}{2}x^T Px + q^T x + r$ be a quadratic function. We want to minimize $f(x)$,

$$\begin{aligned}
 &\min f(x) \\
 &\text{s.t. } Ax = b,
 \end{aligned} \tag{4}$$

where $P \in \mathbb{S}_+^n$ and $A \in \mathbb{R}^{p \times n}$. Note that $\mathbb{S}_+^n$ are $n \times n$ PSD matrices with all eigenvalues ≥ 0 . Then

$$\nabla f(x) = Px + q$$

and

$$\nabla^2 f(x) = P,$$

hence, $\nabla^2 f(x)$ is PSD, which implies that $f(x)$ is convex.

Applying the derived KKT conditions in Eqn. (3) to this special case yields the following optimality conditions:

$$\begin{cases} Px^* + q + A^T \nu^* = 0 \\ Ax^* = b, \end{cases}$$

which implies

$$\begin{cases} Px^* + A^T \nu^* = -q \\ Ax^* + 0 \times \nu^* = b \end{cases} \quad (5)$$

The linear system of equations in Eqn. (5) can be rewritten in the matrix notation:

$$\begin{bmatrix} P & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} x^* \\ \nu^* \end{bmatrix} = \begin{bmatrix} -q \\ b \end{bmatrix}, \quad (6)$$

where

$$\begin{bmatrix} P & A^T \\ A & 0 \end{bmatrix}$$

is called the KKT matrix.

Solving this linear system of equations produces the desired solution to the equality constrained quadratic minimization problem.

4 Solution Existence

There is a unique optimal primal-dual pair (x^*, ν^*) if the KKT matrix is non-singular. If the KKT matrix is singular and the system is solvable, any solution produces an optimal pair (x^*, ν^*) . If the system is unsolvable, the quadratic optimization problem is unbounded below.

Lemma 4.1. *If (6) has no solution, then (4) is unbounded below.*

To prove this lemma, we need another lemma:

Lemma 4.2. *If a matrix $B \in \mathbb{R}^{d \times d}$ is symmetric and $y \notin \text{colspan}(B)$, then $\exists z$ $Bz = 0$ and $\langle z, y \rangle > 0$.*

Proof. Recall: $B = [b_1 \ b_2 \ \dots \ b_d]$, where b_j for $j \in \{1, 2, \dots, d\}$ are column vectors. Then

$$\begin{aligned} \text{colspan}(B) &= \left\{ \sum_{j=1}^d \lambda_j b_j \mid \lambda_j \in \mathbb{R} \right\} \\ &= \{B\lambda \mid \lambda \in \mathbb{R}^d\}. \end{aligned} \quad (7)$$

From Eqn. (7),

$y \notin \text{colspan}(B) \Leftrightarrow B\lambda = y$ has no solution.

Since $y \notin \text{colspan}(B)$, $\exists y_1, y_2$ $y = y_1 + y_2$, where $y_1 \in \text{colspan}(B)$ and $y_2 \perp \text{colspan}(B)$.

In other words, we can decompose the vector y into components parallel and perpendicular to $\text{colspan}(B)$ as in Fig. (1).

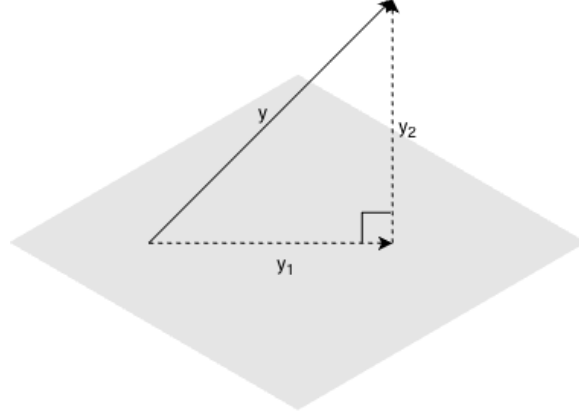


Figure 1: Vector y is decomposed into components y_1 (parallel to the column span of B) and y_2 (perpendicular to the column span of B).

Assume $y_2 = 0$. Then there is no component of y perpendicular to $\text{colspan}(B)$, which means $y = y_1 \in \text{colspan}(B)$. It contradicts the assumption that $y \notin \text{colspan}(B)$. Therefore, $y_2 \neq 0$.

Since $y_2 \perp \text{colspan}(B)$, $\langle y_2, b_i \rangle = 0$ for $i \in \{1, 2, \dots, d\}$.

The equation above can be rewritten as

$$\begin{bmatrix} b_1^T y_2 \\ b_2^T y_2 \\ \vdots \\ b_d^T y_2 \end{bmatrix} = 0 \Leftrightarrow B^T y_2 = 0 \Leftrightarrow B y_2 = 0$$

since B is symmetric.

Now, check that $\langle y_2, y \rangle > 0$.

$$\begin{aligned} \langle y_2, y \rangle &= \langle y_2, y_1 + y_2 \rangle = \langle y_2, y_1 \rangle + \langle y_2, y_2 \rangle \\ &= 0 + \langle y_2, y_2 \rangle \\ &= \|y_2\|^2 > 0, \end{aligned}$$

since $y_2 \neq 0$ and $y_2 \perp y_1 \implies \langle y_2, y_1 \rangle = 0$.

□

Using Lemma 4.2, we can prove Lemma 4.1.

Proof. If (6) has no solution, then

$$\begin{bmatrix} -q \\ b \end{bmatrix} \notin \text{colspan} \begin{bmatrix} P & A^T \\ A & 0 \end{bmatrix}.$$

By Lemma 4.2, $\exists z = \begin{bmatrix} v \\ w \end{bmatrix}$ such that

$$\begin{bmatrix} P & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

and

$$\left\langle \begin{bmatrix} v \\ w \end{bmatrix}, \begin{bmatrix} -q \\ b \end{bmatrix} \right\rangle > 0 \Leftrightarrow -q^T v + b^T w > 0 \quad (8)$$

Let $\hat{x}$ be feasible point of (4), i.e. $\hat{x}$ satisfies the constraint $A\hat{x} = b$.

Then $\hat{x} + tv$ is also feasible for all t , where v is a particular direction and t is the step size.

Indeed,

$$\begin{bmatrix} P & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{cases} Pv + A^T w = 0 \\ Av = 0 \end{cases} \Rightarrow \begin{cases} Pv = -A^T w \\ Av = 0 \end{cases} \quad (9)$$

and

$$A(\hat{x} + tv) = A(\hat{x}) + t(Av) = b + t \times 0 = b,$$

which implies that $\hat{x} + tv$ is feasible.

Then

$$\begin{aligned} f(\hat{x} + tv) &= f(\hat{x}) + t(v^T P \hat{x} + q^T v) + \frac{1}{2} t^2 v^T P v \\ &= f(\hat{x}) + t(\hat{x}^T P v + q^T v) + \frac{1}{2} t^2 v^T P v \\ &= f(\hat{x}) + t(\hat{x}^T (-A^T w) + q^T v) + \frac{1}{2} t^2 v^T (-A^T w) && \text{from (9)} \\ &= f(\hat{x}) + t(-\hat{x}^T A^T w + q^T v) - \frac{1}{2} t^2 w^T A v \\ &= f(\hat{x}) + t(-b^T w + q^T v) - 0 && \text{using } A\hat{x} = b \text{ and } Av = 0 \end{aligned}$$

Since $(-q^T v + b^T w) > 0$ by Eqn. (8), we have $(-b^T w + q^T v) < 0$. Then

$$\lim_{t \rightarrow \infty} f(\hat{x} + tv) = -\infty.$$

Therefore, the quadratic optimization problem is unbounded below. $\square$

References

- [1] S. Boyd, L. Vandenberghe, *Convex Optimization*, Cambridge University Press, 2004.